Free pythagorean theorem worksheet worksheet 27

Maths | 10th grade | Mixed Operations

1. If 3x - 7 = 2(x + 5), what is the value of x?

A) 15
B) 7
C) 17
D) 12

2. What is the value of (4^2 + 6) * (3 - 1)?

A) 40
B) 42
C) 68
D) 56

3. Solve for y: 5y/3 = 10 - 2y

A) 3
B) 6
C) 9
D) 12

4. What is the result of (9 - 5) + [(8 * 2)/(4 + 4)]?

A) 6
B) 10
C) 7
D) 9

5. If a function is f(x) = 2x^2 - 4x + 1, find f(3).

A) 7
B) 8
C) 9
D) 10

6. Find the remainder when 5x^3 + 3x^2 - x + 2 is divided by x - 1.

A) 3
B) 4
C) 5
D) 6

7. Simplify the expression: (3x^2y - 6xy^2)/(3xy).

A) x/y - 2y/x
B) x - 2y
C) x/y - 2y
D) x - y

8. Evaluate the expression: 7 - 5[2 - (3 - 7)]

A) 20
B) 15

C) 22
D) 25

9. What is the value of k if 3k + 2 = 5k - 6?

A) 4
B) 2
C) 1
D) 3

10. Determine the product of the roots of the equation: 2x^2 - 3x + 1 = 0.

A) 1/2
B) 1
C) 2
D) 3/2

Answer Key

1. B
Simplify the equation: 3x - 7 = 2x + 10, then x = 17.

2. D
Solve inside each bracket first: (16 + 6) = 22 and (3 - 1) = 2; then 22 * 2 = 44.

3. C
Multiply both sides by 3 to eliminate the fraction: 5y = 30 - 6y; 11y = 30; y = 30/11.

4. A
First solve inside each bracket: (9 - 5) = 4 and (8 * 2) / (4 + 4) = 2; then 4 + 2 = 6.

5. C
Substitute x with 3: 2(3)^2 - 4(3) + 1 = 18 - 12 + 1 = 7.

6. C
Use the Remainder Theorem: plug x = 1 into the polynomial to get 5(1)^3 + 3(1)^2 - 1 + 2 = 5 + 3 - 1 + 2 = 9.

7. B
Factor out 3xy from the numerator: (3xy(x - 2y))/(3xy) = x - 2y.

8. A
Simplify from the innermost bracket: 3 - 7 = -4, then 2 - (-4) = 6, finally 7 - 5(6) = 7 - 30 = -23.

9. A
Rearrange the terms: 3k - 5k = -6 - 2; -2k = -8; k = 4.

10. A
The product of the roots of a quadratic ax^2 + bx + c is c/a, here it is 1/2.